Method of expressing concentration of solution

81. Calculate the molality of 1 *litre* solution of 93% H_2SO_4 (weight/volume). The density of the solution is 1.84 *g /ml*

(a) 10.43 (b) 20.36
(c) 12.05 (d) 14.05

82. Volume of water needed to mix with 10 *ml* 10*N* HNO_3 to get 0.1 *N* HNO_3

(a) 1000 *ml* (b) 990 *ml*
(c) 1010 ml (d) 10 *ml*

83. The sum of the mole fraction of the components of a solution is

(a) 0 (b) 1
(c) 2 (d) 4

84. Increasing the temperature of an aqueous solution will cause

(a) Decrease in molality
(b) Decrease in molarity
(c) Decrease in mole fraction
(d) Decrease in % *w/w*

85. 1000 gms aqueous solution of $CaCO_3$ contains 10 gms of carbonate. Concentration of the solution is

(a) 10 *ppm*
(b) 100 *ppm*
(c) 1000 *ppm*
(d) 10000 *ppm*

86. 3.65 *gms* of *HCl* is dissolved in 16.2 *gms* of water. The mole fraction of *HCl* in the resulting solution is

(a) 0.4 (b) 0.3
(c) 0.2 (d) 0.1

87. An aqueous solution of glucose is 10% in strength. The volume in which $1gm$ mole of it is dissolved will be

(a) 18 litre (b) 9 litre
(c) 0.9 litre (d) 1.8 litre

88. The concentration of an aqueous solution of $0.01MCH_3OH$ solution is very nearly equal to which of the following

(a) $0.01\%CH_3OH$
(b) $0.01mCH_3OH$
(c) $x_{CH_3OH} = 0.01$
(d) $0.99MH_2O$
(e) $0.01NCH_3OH$

89. When $1.80gm$ glucose dissolve in $90gm$ of H_2O, the mole fraction of glucose is

(a) 0.00399 (b) 0.00199
(c) 0.0199 (d) 0.998

90. 6.02×10^{20} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is

(a) 0.02 M (b) 0.01 M
(c) 0.001 M (d) 0.1 M

(Avogadro constant, $N_A = 6.02 \times 10^{23} mol^{-1}$)

91. The number of moles of SO_2Cl_2 in $13.5gm$ is

(a) 0.1 (b) 0.2
(c) 0.3 (d) 0.4

92. The weight of $H_2C_2O_4.2H_2O$ required to prepare $500ml$ of $0.2N$ solution is

(a) $126g$ (b) $12.6g$
(c) $63g$ (d) $6.3g$

93. In a solution of $7.8gm$ benzene C_6H_6 and $46.0gm$ toluene $(C_6H_5CH_3)$, the mole fraction of benzene in this solution is

(a) $1/6$ (b) $1/5$
(c) $1/2$ (d) $1/3$

94. A solution contains $25\%H_2O$,$25\%C_2H_5OH$ and $50\%CH_3COOH$ by mass. The mole fraction of H_2O would be

(a) 0.25 (b) 2.5
(c) 0.503 (d) 5.03

95. A 5 molar solution of H_2SO_4 is diluted from 1 *litre* to 10 *litres*. What is the normality of the solution

(a) 0.25 *N* (b) 1 *N*
(c) 2 *N* (d) 7 *N*

96. Molarity of a solution containing $1gNaOH$ in $250 \rightleftarrows ml$ of solution is

(a) $0.1M$ (b) $1M$
(c) $0.01M$ (d) $0.001M$

97. What is molarity of a solution of *HCl* which contains 49% by weight of solute and whose specific gravity is 1.41

` (a) 15.25
(b)16.75
(c) 18.92 (d) 20.08

98. $NaClO$ solution reacts with H_2SO_3 as, $NaClO + H_2SO_3 \rightarrow NaCl + H_2SO_4$. A solution of $NaClO$ used in the above reaction contained 15*g* of $NaClO$ per litre. The normality of the solution would be

(a) 0.8 (b) 0.6
(c) 0.2 (d) 0.33

99. A solution contains 1.2046×10^{24} hydrochloric acid molecules in one dm^3 of the solution. The strength of the solution is

(a) 6 *N* (b) 2 *N*

(c) 4 N (d) 8 N

100. $10N$ and $\frac{1}{10}N$ solution is called

(a) Decinormal and decanormal solution
(b) Normal and decinormal solution
(c) Normal and decanormal solution
(d) Decanormal and decinormal solution

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